PAPER ID-410297

Roll No: | | | | | | | | | | | | | |

# BTECH
## (SEM VI) THEORY EXAMINATION 2023-24
## TRANSPORTATION ENGINEERING

**TIME: 3 HRS** **M.MARKS: 100**

**Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.

### SECTION A

**1. Attempt *all* questions in brief.**

| | | |
|---|---|---|
| a. | Write the major modes of transportation. | 2 |
| b. | Write the features of the third 20 years (Lucknow Road Plan) (1981-2001). | 2 |
| c. | What do you mean by geometric design? | 2 |
| d. | What are the cross-sectional elements of highway geometric design? | 2 |
| e. | List the type of road user characteristics. | 2 |
| f. | Determine the theoretical capacity of a traffic lane having one-way traffic low at a stream speed of 40 kmph. Assume the average space gap between vehicles to follow the relation Sg=0.278 Vt. V is the stream speed in kmph and t is the average reaction times=0.7 seconds and assume average length of vehicles = 5.0 m. | 2 |
| g. | Write the different types of emulsion. | 2 |
| h. | What are the constituents of mix? | 2 |
| i. | Write the types of highway construction. | 2 |
| j. | What do you mean by dry lean concrete? | 2 |

### SECTION B

**2. Attempt any *three* of the following:**

| | | |
|---|---|---|
| a. | Briefly explain the engineering surveys needed for a highway project and the data to be collected. | 10 |
| b. | A parabolic camber is provided on a two-lane road of width 7.5 m in a region of heavy rainfall with thin bituminous surface. If it is proposed to construct the road with Cement concrete. Find the change in the height of the crown. | 10 |
| c. | Explain with a diagram effect of mounting and overhang on the shadow of the vehicle. | 10 |
| d. | Determine the unit weight of a bituminous mix containing 70% coarse aggregate, 24% fine aggregate, and 6% bitumen by weight of the mixture. The air voids after compaction are 8%. The specific gravities of the material are as under:<br>Sp. Gravity of coarse aggregate =2.80, Sp. The gravity of fine aggregate=2.66, Sp. Gravity of bitumen=1.00 | 10 |
| e. | Describe specifications of materials for water-bound macadam roads. | 10 |

### SECTION C

**3. Attempt any *one* part of the following:**

| | | |
|---|---|---|
| a. | Explain with sketches the various factors affecting the highway alignment, | 10 |
| b. | Write the recommendations made by the Jayakar committee and gives the accepted recommendations of the Jayakar committee. | 10 |

**4. Attempt any *one* part of the following:**

| | | |
|---|---|---|
| a. | On a two-way traffic road, the speed of overtaking vehicles is 100 kmph and 50 kmph. If the average acceleration is 0.92 m/s$^2$. Determine the overtaking distance. Draw a neat sketch of the overtaking zone and show the positions of the sign posts. | 10 |
| b. | Discus gradients and its type. Specify the values recommended by IRC for plains and hills. | 10 |

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**5. Attempt any *one* part of the following:**

| | | |
|---|---|---|
| a. | Based on data for spot studies given in table. Calculate upper and lower speed limit regulation as well as speed for design. | 10 |

| Speed Range (km/hr.) | Number of vehicles |
|---|---|
| 0-10 | 12 |
| 10-20 | 18 |
| 20-30 | 68 |
| 30-40 | 90 |
| 40-50 | 207 |
| 50-60 | 252 |
| 60-70 | 21 |
| 70-80 | 44 |
| 80-90 | 32 |
| 90-100 | 9 |

| | | |
|---|---|---|
| b. | Design two phases traffic signals by Webster's method using the following data. Take all red-time required for pedestrian crossing is 12 sec. and amber time of 2 sec. for each lane. | 10 |

| Road | Average Normal Flow (in vehicle/hr.) | Saturation Flow (in vehicle/hr.) |
|---|---|---|
| A | 400 | 1250 |
| B | 250 | 100 |

**6. Attempt any *one* part of the following:**

| | | |
|---|---|---|
| a. | Explain desirable properties of road aggregates. The weight of coarse aggregate having specific gravity 2.65, which is filled into a cylinder of volume 0.003 m3 is 5247 gm. What is angularity number? | 10 |
| b. | Explain with graphical representation of Equivalent Single Wheel Load stress concept. | 10 |

**7. Attempt any *one* part of the following:**

| | | |
|---|---|---|
| a. | Describe step wise construction of cement concrete pavement. | 10 |
| b. | Write advantages and disadvantages of following; (i) Gravel Road (ii)Bituminous Road. | 10 |

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